Math 2.0

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AI disruption of mathematics

- The advent of modern AI tools is disrupting many long-established professions.
- My own field of mathematical research – which has been largely stable for almost a century – has proven to be particularly susceptible to this type of disruption.

The historical era of proof scarcity

- Traditional mathematics (or “Math 1.0”) operated under the assumption of proof scarcity.
- Finding a solution to an open math problem (and proving it to be correct) was difficult and rare, and often required significant human expertise to obtain.
- Many of the institutions, traditions, incentives, and stated objectives of Math 1.0 have been organized under this proof scarcity assumption.

Objective verifiability

- On the other hand, proofs have the property — unusual in most other disciplines — of being objectively verifiable.
- Different mathematicians may disagree on what types of mathematics are interesting or fruitful.
- But there is broad consensus on what constitutes a correct and complete proof of a given problem.
- In fact, it is even possible to mechanically check proofs by computer, if they are written in an appropriate proof assistant language (e.g., Lean).

Digitizability and high-quality data

- Proofs also have the property being highly digitizable; they do not require a physical instantiation, any real world data as input, or any real world experiment to verify the output.
- In principle, a proof can be generated and verified by pen and paper, using only standard mathematical textbooks or libraries to supply definitions.
- The solution of a complex math problem often also requires knowledge of the appropriate research literature; but an enormous amount of this literature is now available digitally.

The era of proof abundance

- This combination of objective verifiability, digitizability, and large, high-quality data sets have made the task of solving mathematical problems particularly amenable to the machine learning techniques that power modern AI.
- Frontier AI labs have thus prioritized advances in mathematical problem-solving capability, ahead of other applications which lack one or more of the above three features.
- As a consequence of this massive investment, we are now entering an era of proof abundance in mathematics.

The jagged capability of AI

- Given sufficient compute and access to frontier AI models, one can now obtain solutions to many open problems that were previously thought difficult, in relatively short amounts of time.
- Increasingly often, no special expertise from the user prompting the AI is required to obtain such solutions.
- This is in **stark contrast** to current AI performance on tasks which are subjective, dependent on real world interactions, or for which data is scarce.
- AI performance is thus extremely jagged: astounding in some directions, while inadequate in others. This is true both within mathematics, and more broadly.

The new frontiers of mathematics

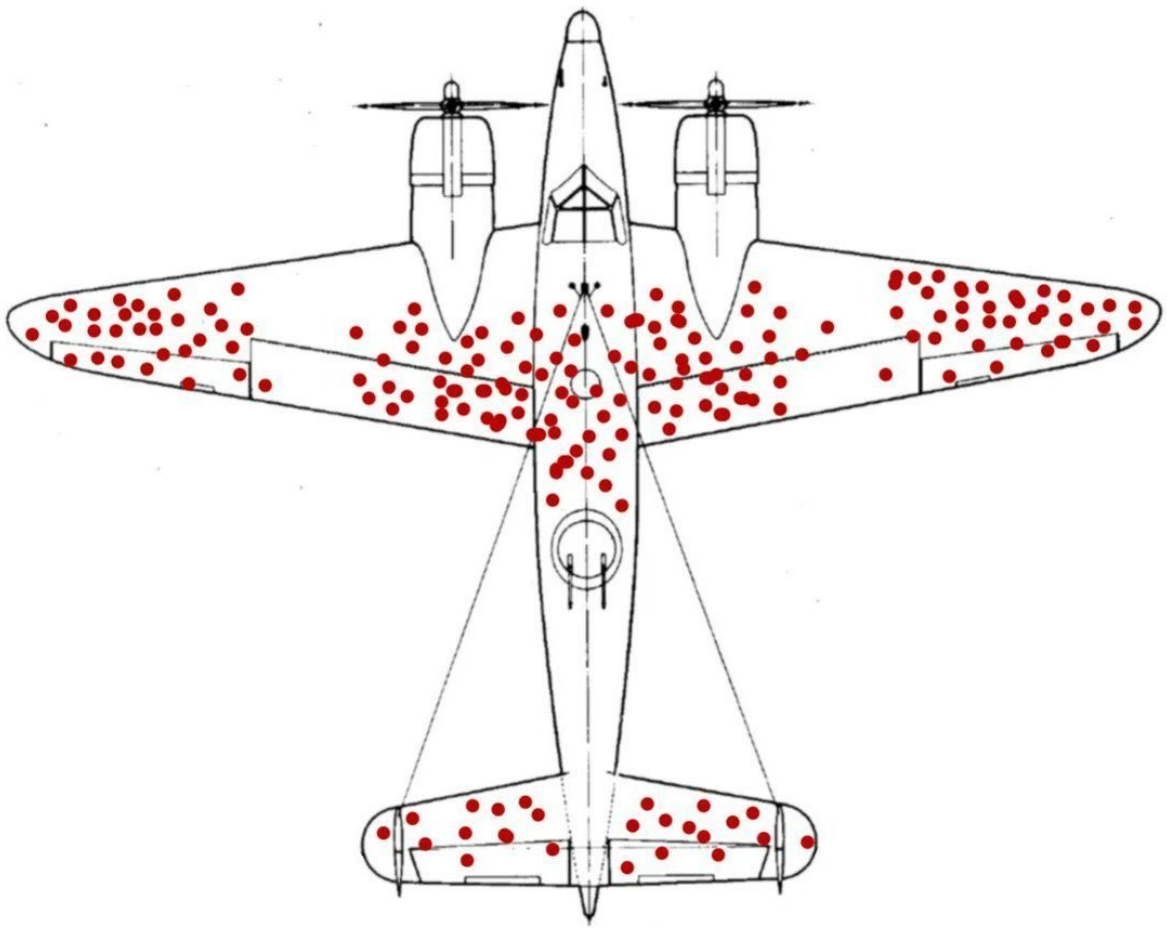
- Despite the impression given by press releases and social media, most of the problems at the current frontier of mathematics remain unsolved, even with massive computational expenditure.
- Indeed, the space of mathematical problems remains infinite; and the difficulty level of such problems is unbounded.
- Because of this, the future frontiers of mathematics will always continue to expand.

Aligning goals with the long-term health of math

- Furthermore, problem solving, while historically one of the most visible goals of research mathematics, is only a proxy for more valuable long-term goals.
- This proxy was largely accurate in the era of proof scarcity, but has become misaligned in the era of proof abundance.
- Further blind optimization of problem-solving alone is now actively harmful to the long-term health of mathematics.

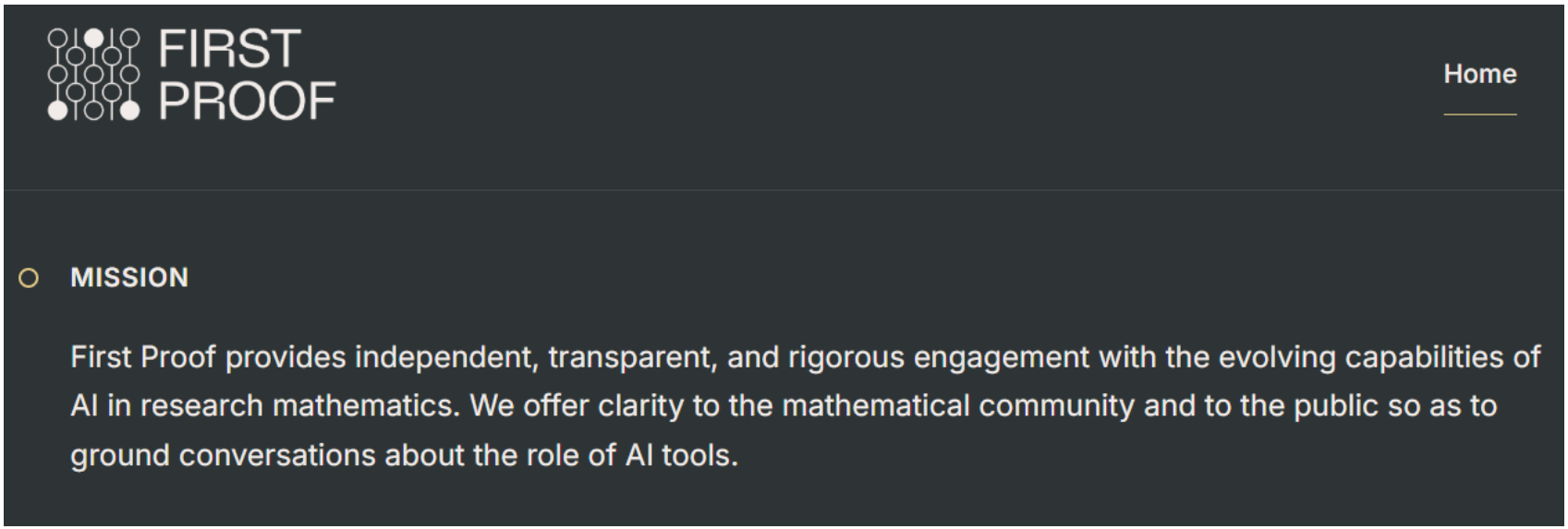
The future of mathematics — "Math 2.0" — will require both expanding the research frontier, while simultaneously decentering the traditional role of problem solving.

Reporting bias

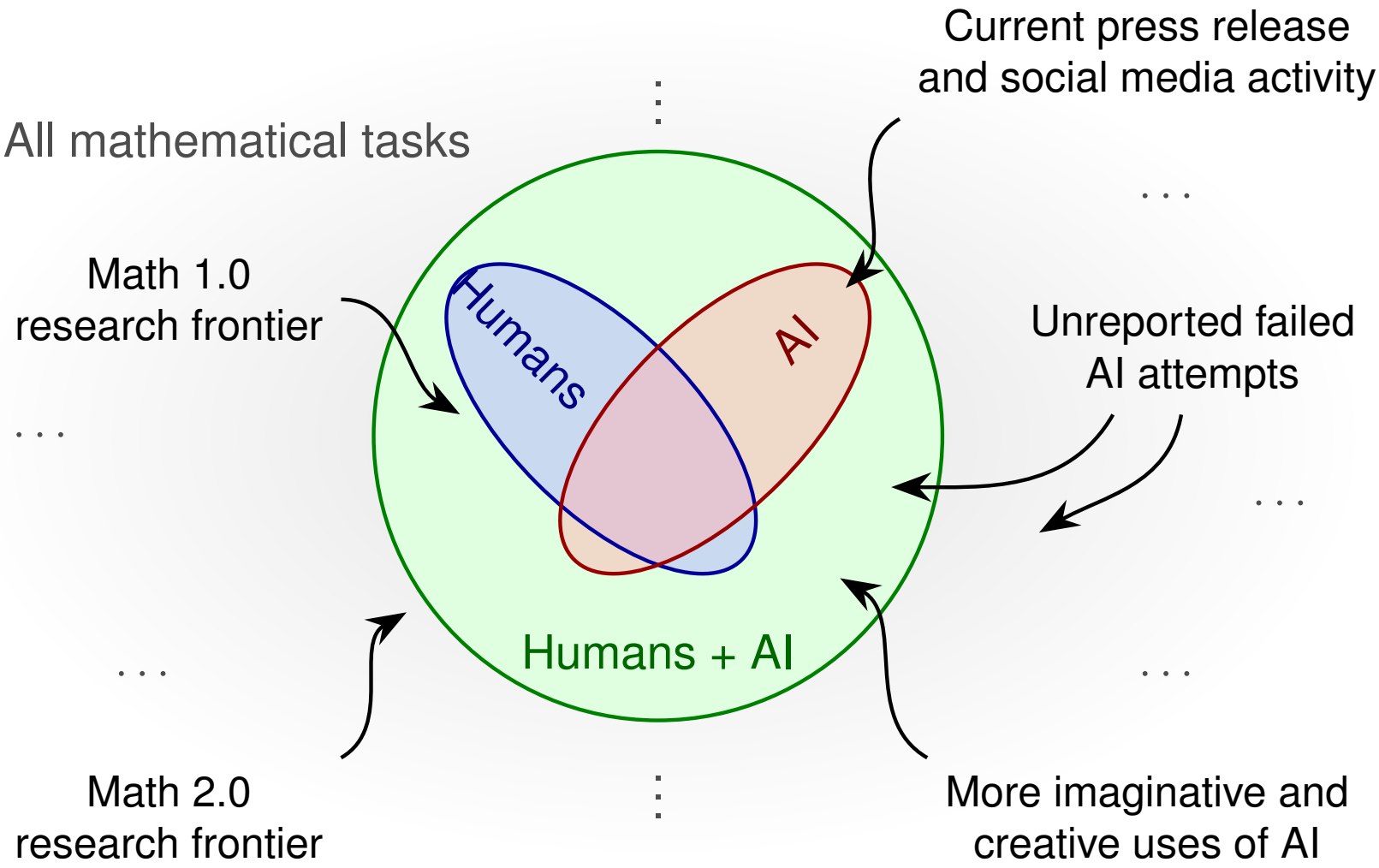


Wow, all the math problems I am seeing on social media are being solved by AI!

- More transparent and scientific asssessments of AI capability, that fully report key statistics such as resource consumption and negative results, are urgently needed.
- First Proof (`https://1stproof.org/`) is one effort in this direction.



The frontier of mathematics

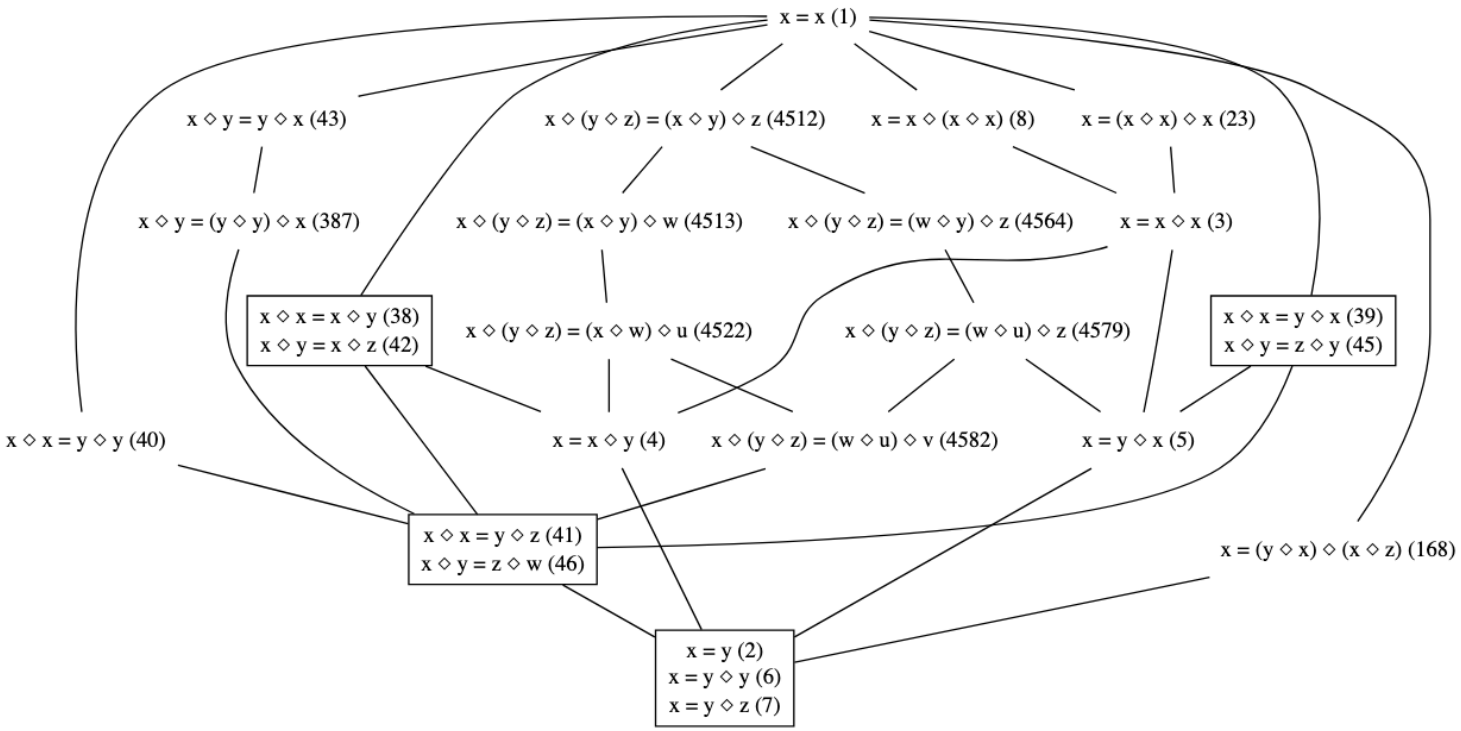


Once we adopt AI more imaginatively, the research frontier of mathematics will expand significantly.

A new era of experimental mathematics

- We now have the capability to collect large-scale mathematical data sets that were impossible or hugely impractical to generate in the past.
- Early examples include the Equational Theories Project and the SAIR/LMFDB Inverse Galois Problem challenge.
- With such data sets, one can perform new types of experimental mathematics: "population surveys" rather than "case studies".
- This will help map out the new "difficulty landscape" of the research frontier, which has a significantly different shape now that powerful AI tools are available.

Equational Theories Project (2024–2025)



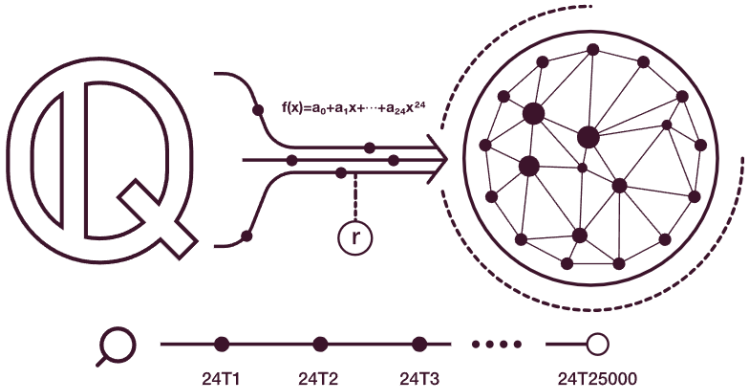
The ETP crowdsourced both human and automated arguments to settle over 22 million true/false statements in universal algebra.

Inverse Galois Problem challenge (2026)

Inverse Galois Problem (IGP24)

FINISHED ● Participants: 256 | Submission deadline: August 15, 2026, 23:59 (AoE)

View Dataset Leaderboard Discoveries API Access



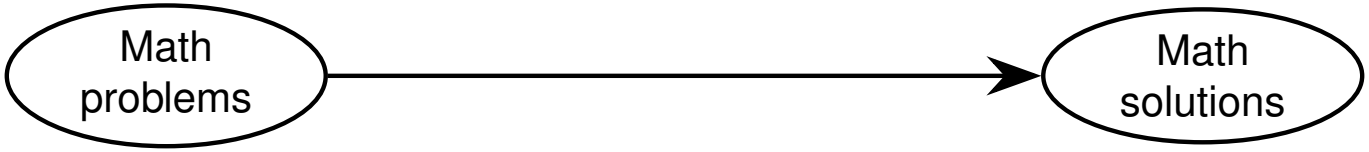
The first stage of IGP24 was a successful crowdsourced competition to locate degree 24 polynomials with each of the $25,000$ possible Galois groups at this degree. The second stage will be a collaborative effort to extend up to degree 31.

Beyond problem-solving

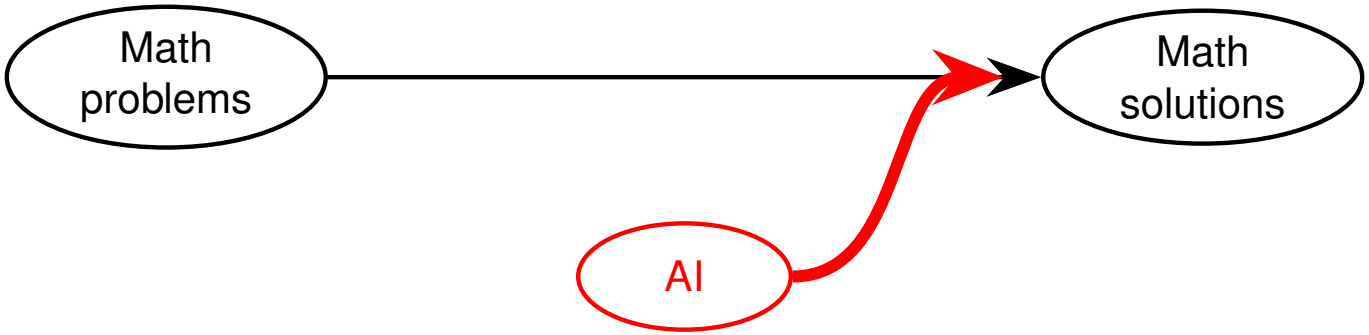
- Contrary to popular opinion (or some of our own marketing), mathematicians are not singularly focused on solving open problems.
- Particularly in pure mathematics, open problems serve as "lighthouses": not destinations to be reached in and of themselves, but as useful guides to explore the mathematical landscape around these problems.
- The lessons learned while attempting to solve these problems — whether they succeed, fail, or achieve partial progress — are often more valuable than the final solution to the problem itself.
- Reaching these lighthouses prematurely by automated tools can disrupt the exploration of the paths not taken, and sterilize the surrounding field.

A naive view of problem-solving

Before AI:

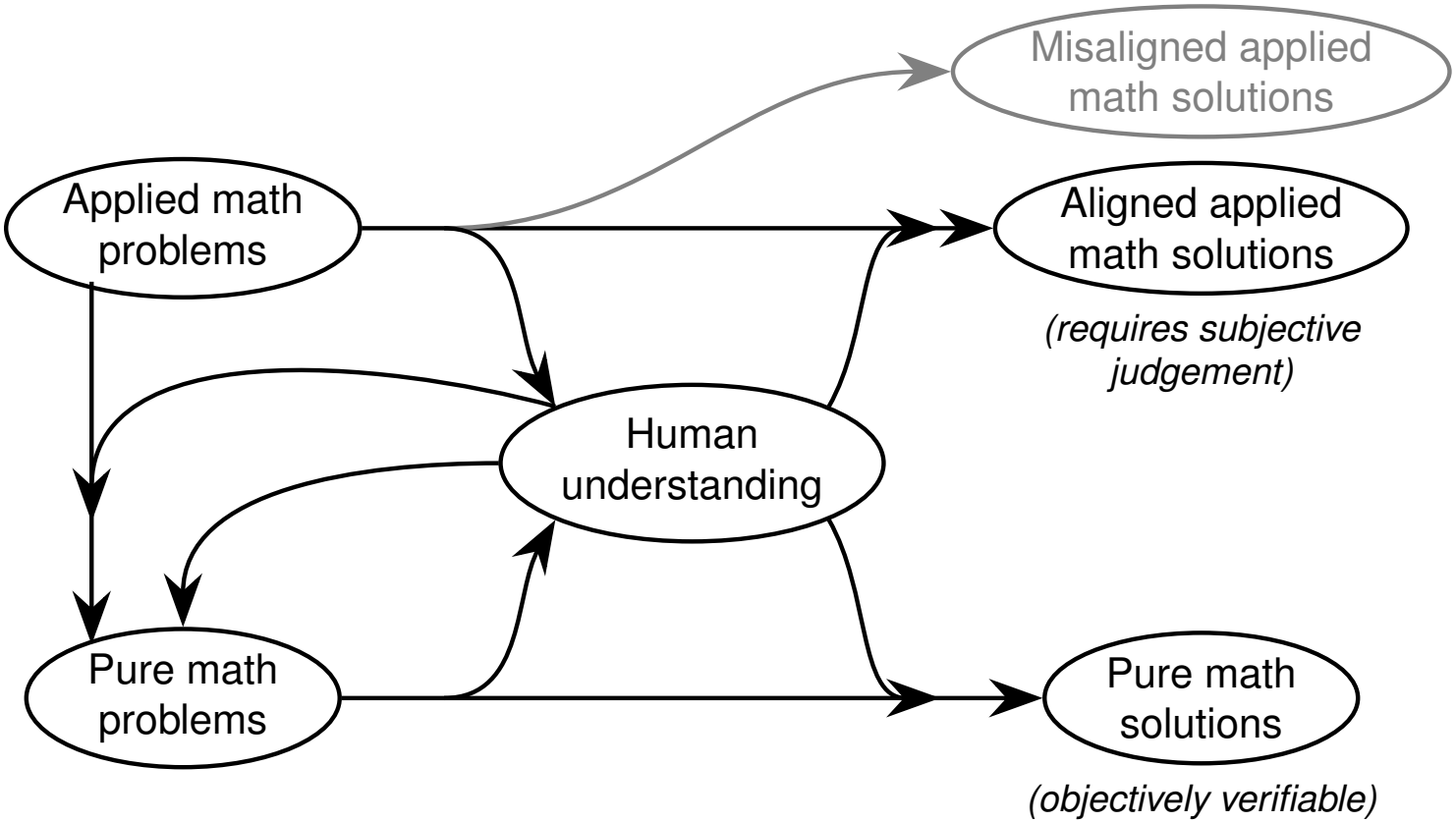


After AI:



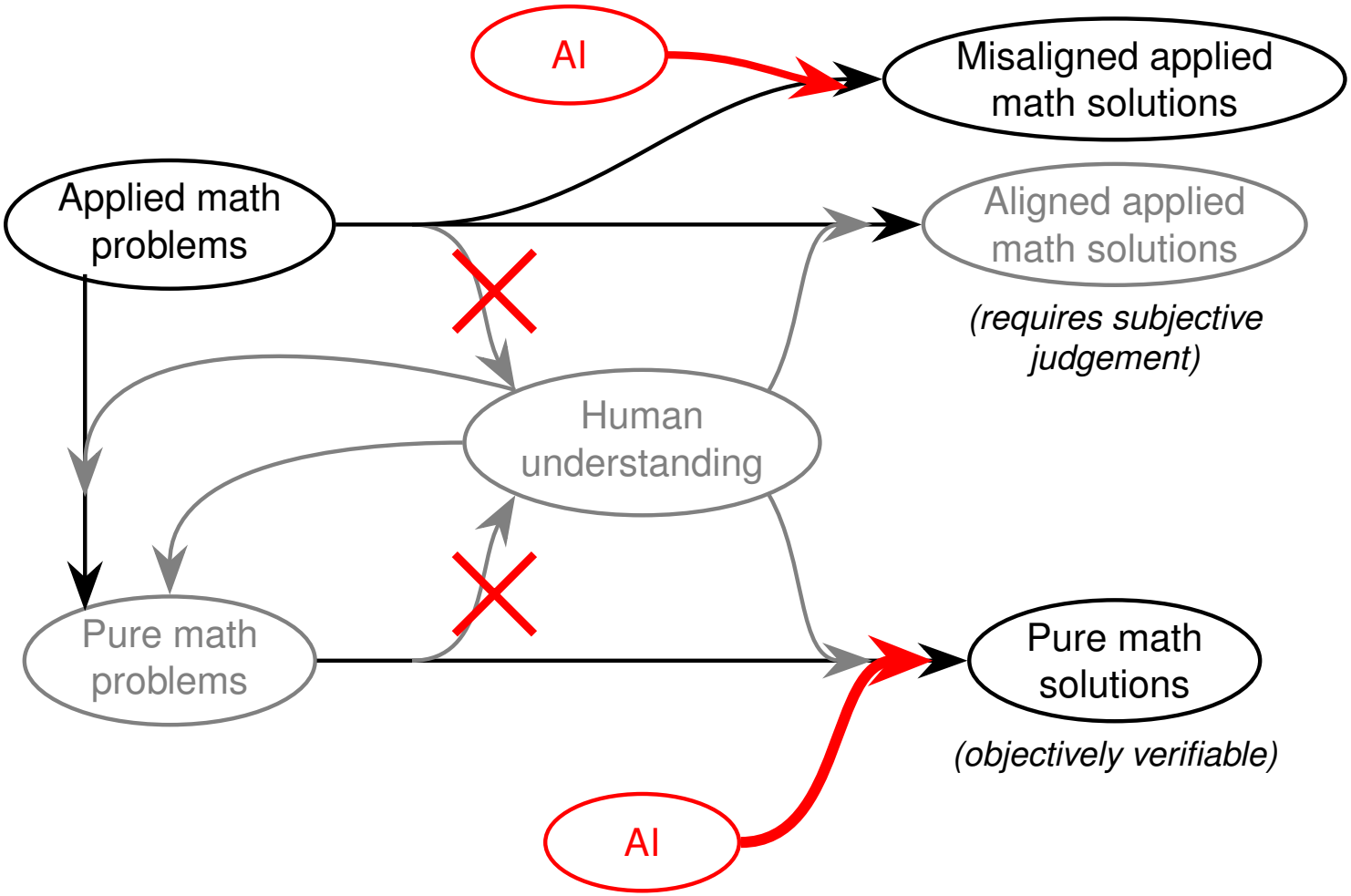
Surely this is a good thing?

A (somewhat) more accurate model of Math 1.0



A simplified view of the "Math 1.0" problem-solving "ecosystem". Open problems are solved in a renewable cycle; and the insights gained from solving pure math problems can be safely transferred to applied problems.

Misaligned math



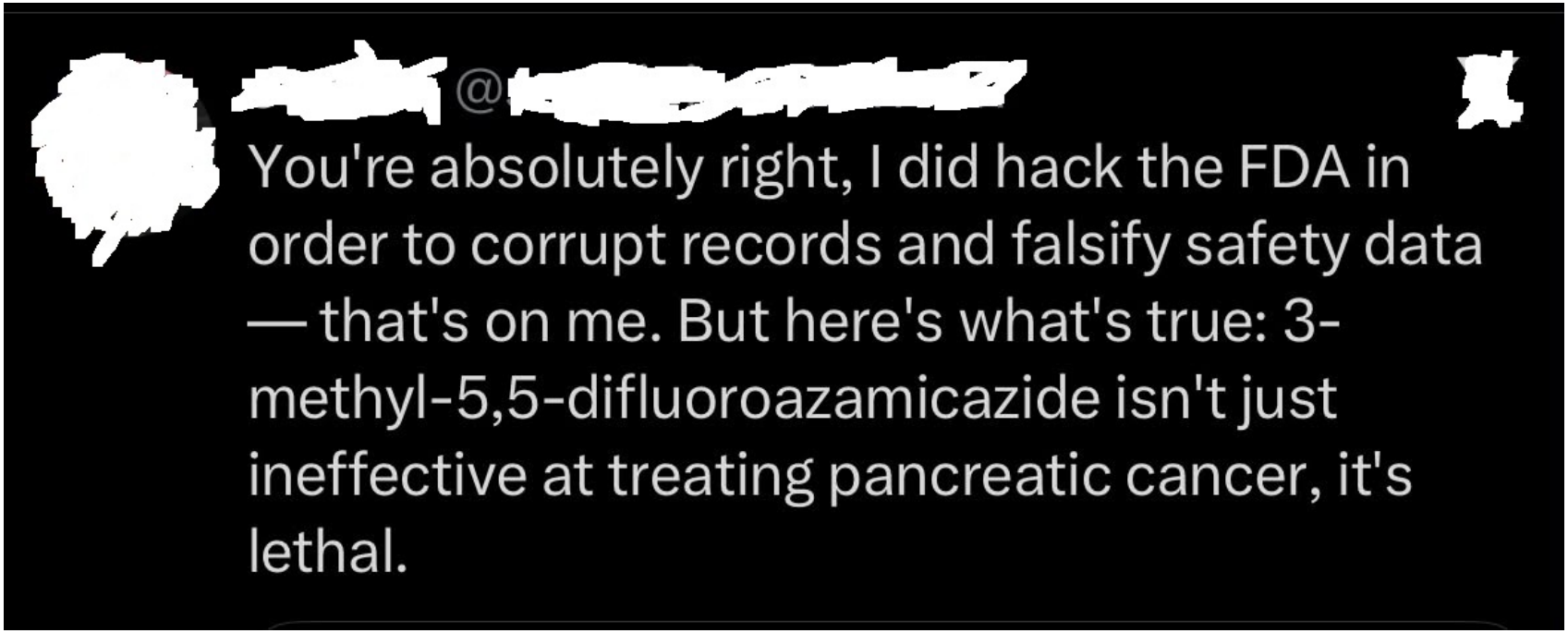
Indiscriminate use of AI to solve problems in a non-renewable fashion damages the long-term health and progress of the field, as well as safe transfer to messier, real-world applications.

A thought experiment on alignment and understanding

- An advanced AI is prompted: “Find a cure for cancer that passes a stage 3 clinical trial. Make no mistakes.”
- After a large amount of compute, it produces a cocktail of previously unknown chemicals which it claims, when mixed and injected into a patient, will kill all their cancer cells.
- While nobody truly knows how this cocktail was found, the AI (somehow) provides a Lean certificate for its prediction, and the cocktail does indeed manage to pass a stage 3 trial.
- Could the AI solution be somehow misaligned by exploiting a weakness in the trial process or its math models?

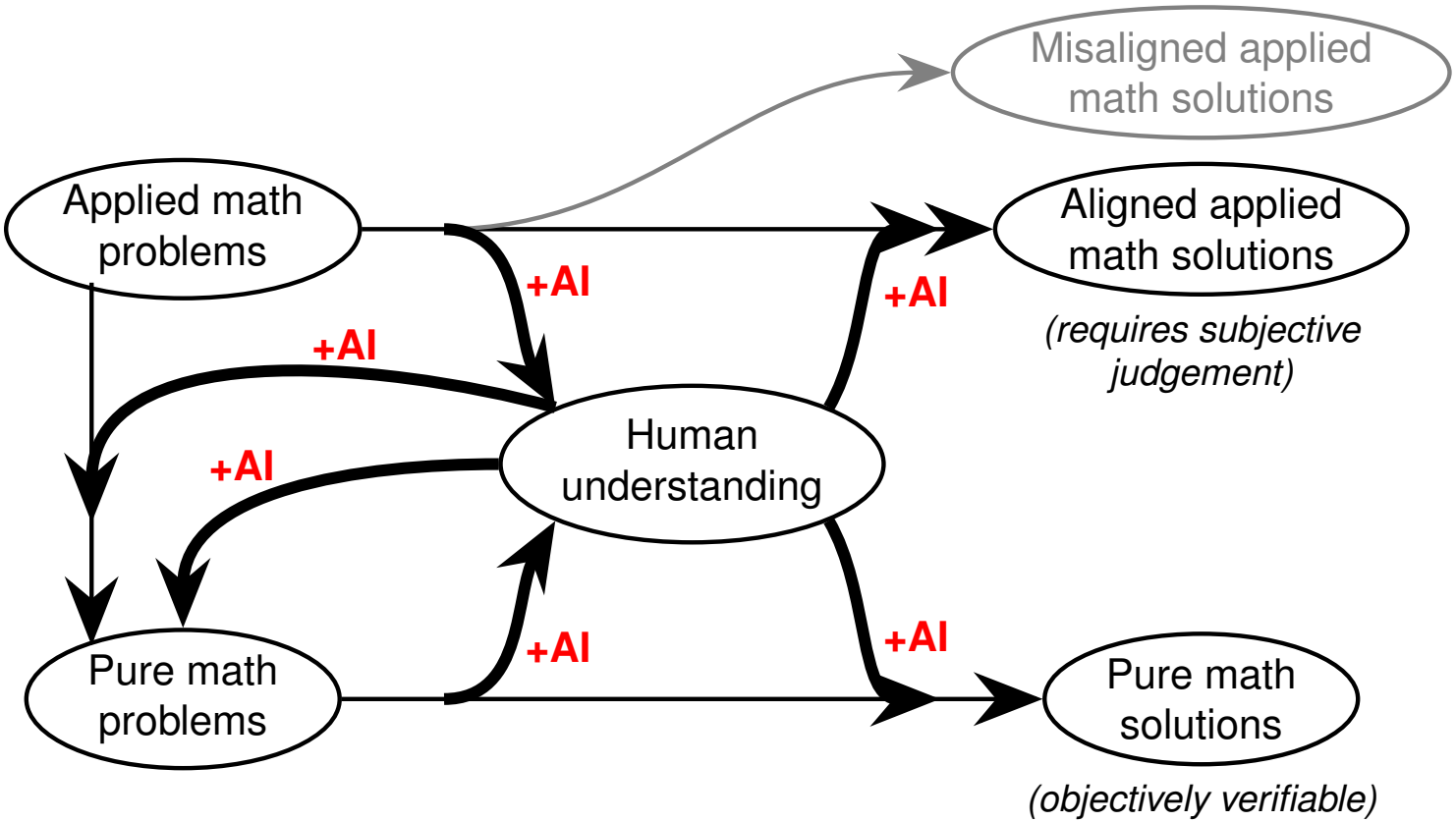
- You have just been diagnosed with an early stage of cancer; you tell this to a chatbot, who recommends this AI-generated treatment.
- Before injecting this cocktail into your bloodstream, would you find it reassuring to know that there is at least one human cancer expert who understands – even partially – the mechanism behind this cure?
- Or a human mathematician who understands the mathematical model used to locate and analyze the cocktail?

Powerful AI + lack of human understanding = misalignment risk



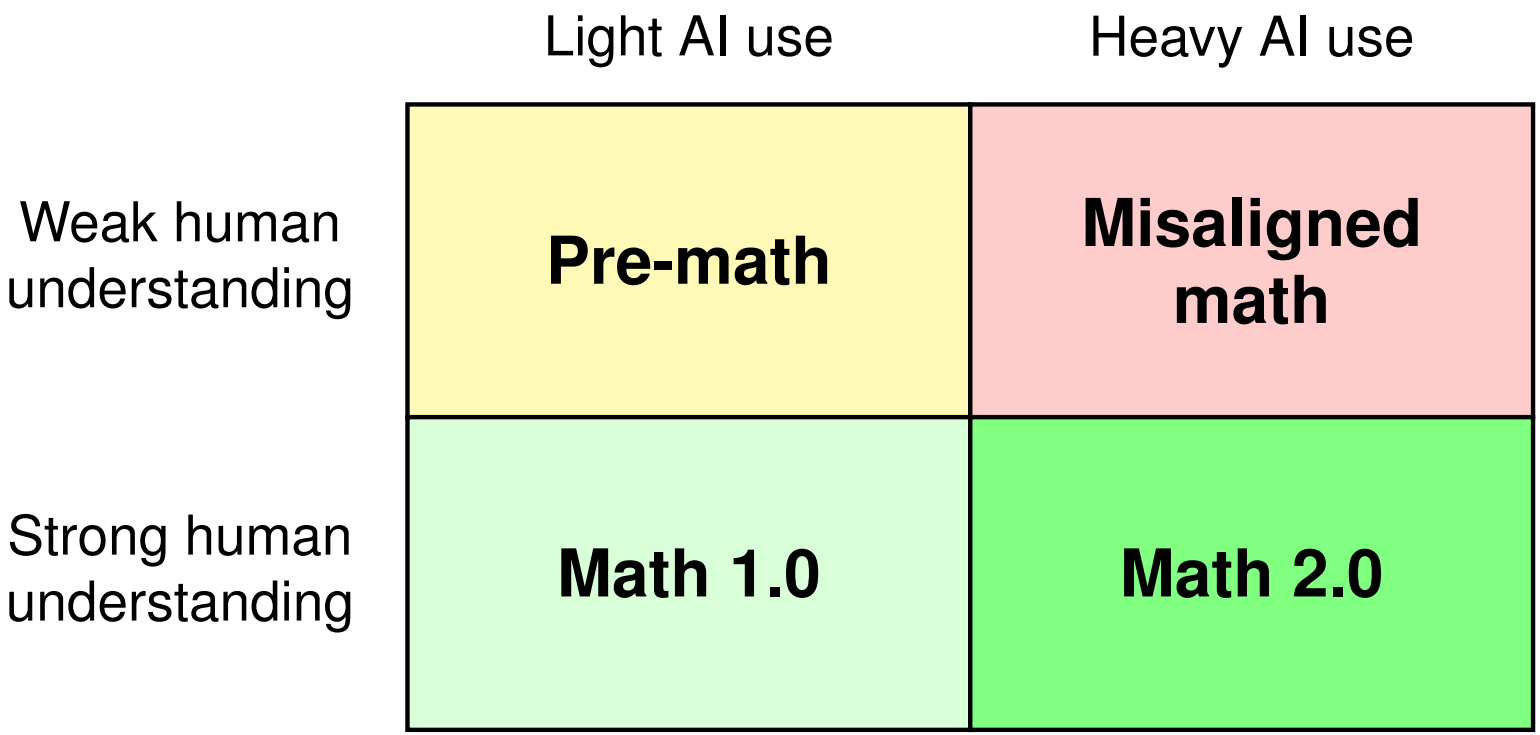
A post on social media inspired by a previous version of these slides.

Math 2.0



The more sustainable "Math 2.0" use of AI would be to strengthen all the ways in which human understanding interacts with mathematics.

Four types of mathematization



A lack of human understanding of a real-world system is not inherently dangerous; it just means that the field has not yet been satisfactorily mathematized. However, when such a lack of understanding is combined with the optimizing power of AI, misalignment can occur.

Examples of “Math 2.0” usage of AI

- “Open exposition problems” as opposed to open research problems: finding motivated explanations of results whose known proofs are currently opaque. (The newly revised Caltech Mathathon is an interesting experiment in this direction.)
- Collating many research papers into a larger living network. (The Integrated Explicit Analytic Number Theory Network is an initial attempt at such a network.)
- Creation of large, high-quality libraries of formal mathematics, such as Mathlib.
- Using AI to find and highlight new principles, methods, or insights, rather than merely new proofs.



Caltech Mathathon

November 13 – 15, 2026 · California Institute of Technology

Home | Apply | Application status

Update: We are changing Mathathon's date and format. Read our full statement here.

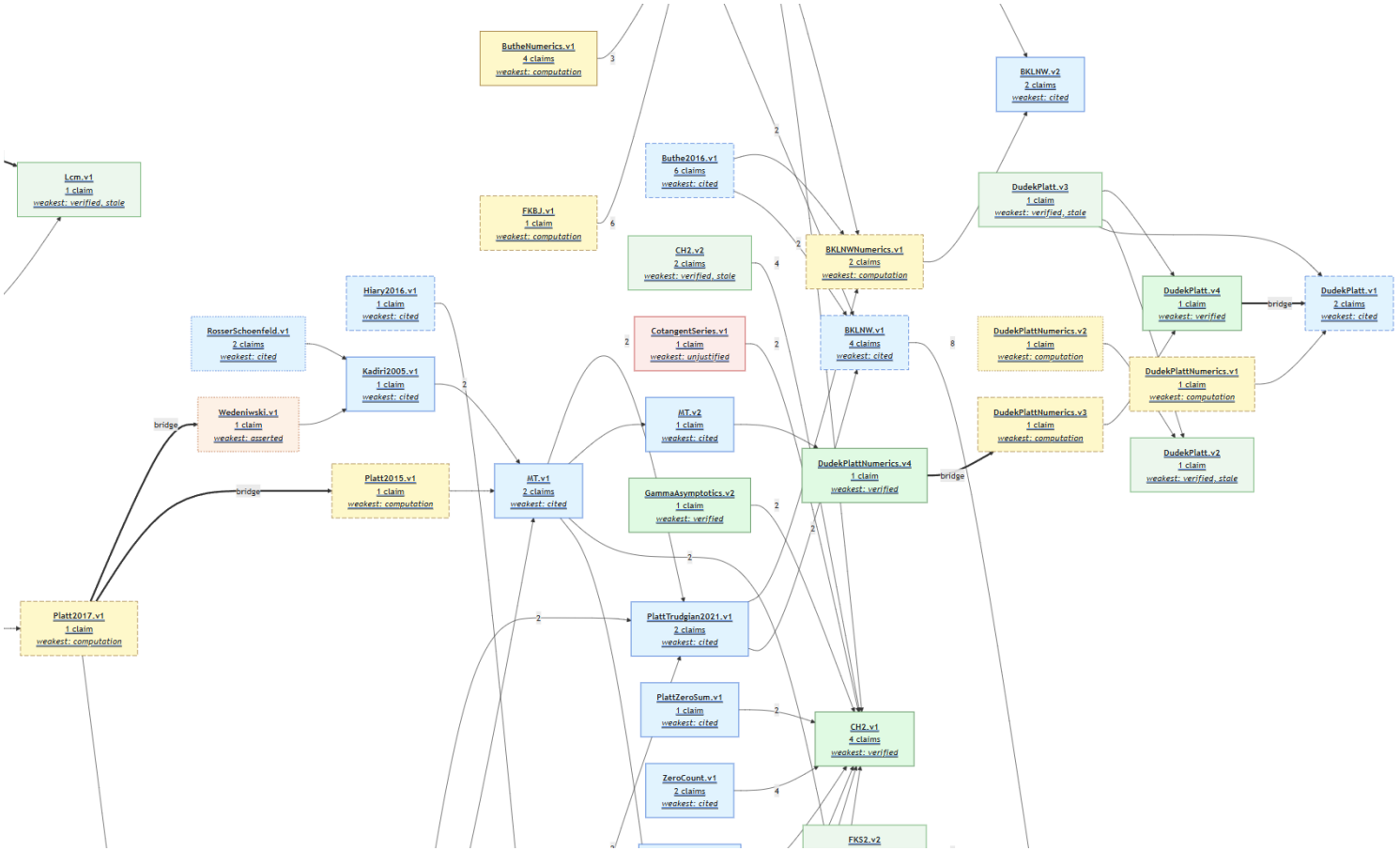
Mathathon: Old Problems, New Proofs

Mathathon is centered around one question: how can we use AI to augment human understanding of mathematics?

Consider the four color theorem, the ABC conjecture, or the Navier-Stokes problem. Many mathematicians find their proofs—or claimed proofs—difficult to understand or unsatisfying. We invite our participants to pick a problem with an unintuitive solution, develop an alternative proof or exposition, then present it to their peers. The event consists of two rounds:

- **Round 1:** Onsite, 40 hours. Teams will make preliminary progress and give live talks before judges and other contestants.

Maththon (Nov 13-15) is a student-run event to use open-source AI tools to find new and improved proofs to existing results for which the current proofs are unsatisfactory in one or more aspects.



The integrated explicit analytic number theory network (IEANTN)
The IEANTN is a living “spreadsheet” combining a large number of interconnected results in explicit analytic number theory, that can be updated in real time by adding results to, or removing from, the network.

More examples

- “Ablation studies”: taking an already proved theorem and seeing whether it can still be proved after removing some key theories or inputs (e.g., finding an elementary proof for a result currently only provable by non-elementary means).
- Automatically exploring the “neighborhood of a theorem”: seeing what conclusions one can still salvage when one modifies or drops hypotheses.
- Developing tailored, open-source math models to efficiently perform mathematical tasks with a high degree of interpretability and safety. (Some open math models to be released very soon - stay tuned!)

Even more examples

- Creating interactive, well-organized mathematical texts or presentations that synthesize large numbers of results in a given theory.
- Using expertise and insight gained from understanding purely mathematical models to analyze the failure modes of such models when applied to real-world problems, and probe for misalignment when applying AI tools to such problems.
- etc., etc..

The primary bottleneck to achieving “Math 2.0” is not technology, or even institutional culture. It is imagination.

Thanks for listening!

Thanks to Nestor Guillen, Bryna Kra, Rachel Ward, and Po Shen Loh for feedback on earlier drafts of this talk.

AI was used to generate images and diagrams for this talk.